



Faculty of Science
Faculty of Economics & Business

ADVANCED ANALYTICS IN BUSINESS

Course report

by

Vince COPPENS (r0932096), Joren RENDERS (r0948114),
Thomas PAESSCHIERSENS (r0934831), Flor VAN AKEN (r0955537),
Vidar ELFVING (r0941939), Matthias GELDHOF (r0933287)

Professor S. VANDEN BROUCKE

This report details the implementation and results of four specialized assignments, demonstrating the practical application of theoretical concepts in advanced analytics to complex business challenges.

Academic year 2025–2026
First Master of Business and Information Systems Engineering
First Master of Statistics and Data Science
First Master of Mathematics
Catholic University Leuven

Contents

1	Example chapter	1
1.1	Chapter title	1
	Bibliography	3

1. Example chapter

1.1 Chapter title

This is an *example chapter*. Use `\mainchapter{Your title}` in `main.tex` for the numbered chapter heading (what you see as “1. ...” in the PDF and table of contents). Inside the chapter file, use `\section` and `\subsection` for structure.

1.1.1 Subtitle (section and subsection)

`\section{...}` is a major heading. `\subsection{...}` is a subtitle under it. You can add normal paragraphs like this one: write sentences, use **bold**, *italics*, and `typewriter` for code or identifiers.

1.1.2 Bullet list

Use an `itemize` environment for bullet lists:

- First item
- Second item
- Third item (add as many as you need)

1.1.3 Figure: importing an image

Put image files under `report/figures/` (or extend `\graphicspath` in `main.tex`). Reference paths without the folder name if that folder is on the graphic path.



Figure 1.1: Example figure: include an image with `\includegraphics`.

Figure 1.1 shows a small example. Change `width` to scale the image.

1.1.4 Citing a source

Cite entries from `references.bib` with `\cite{key}`. Example: work in machine learning has a long history [1]. Web pages and reports often use a `@misc` entry [2].

1.1.5 Source code listing

Short code snippets can use the `lstlisting` environment (configured in `main.tex`):

Listing 1.1: Example Python source.

```
# Example: load data
import pandas as pd

df = pd.read_csv("data.csv")
print(df.head())
```

For inline code, use `\texttt{...}` or `print(x)`.

Bibliography

- [1] A. L. Samuel, “Some studies in machine learning using the game of checkers,” *IBM Journal of Research and Development*, vol. 3, no. 3, pp. 211–229, 1959.
- [2] KU Leuven, “KU Leuven (example web source).” <https://www.kuleuven.be/>, 2026. Example misc entry; replace with real sources for your report.